

Ergonomics

- The study of people's efficiency in their working environment.
- Human factors and ergonomics (commonly referred to as human factors) is the application of psychological and physiological principles to the engineering and design of products, processes, and systems.
- The goal of human factors is to reduce human error, increase productivity, and enhance safety and comfort with a specific focus on the interaction between the human and the thing of interest.
- In research, human factors employs the scientific method to study human behaviour so that the resultant data may be applied to the four primary goals.
- In essence, it is the study of designing equipment, devices and processes that fit the human body and its [cognitive](#) abilities.
- The two terms "human factors" and "ergonomics" are essentially synonymous.

... Ergonomics

- Human factors is employed to fulfill the goals of [occupational health and safety](#) and [productivity](#). It is relevant in the design of such things as safe furniture and easy-to-use interfaces to machines and equipment.
- Proper ergonomic design is necessary to prevent [repetitive strain injuries](#) and other [musculoskeletal disorders](#), which can develop over time and can lead to long-term disability.
- Human factors and ergonomics is concerned with the "fit" between the user, equipment, and environment or "fitting a job to a person"

Fields of Ergonomics

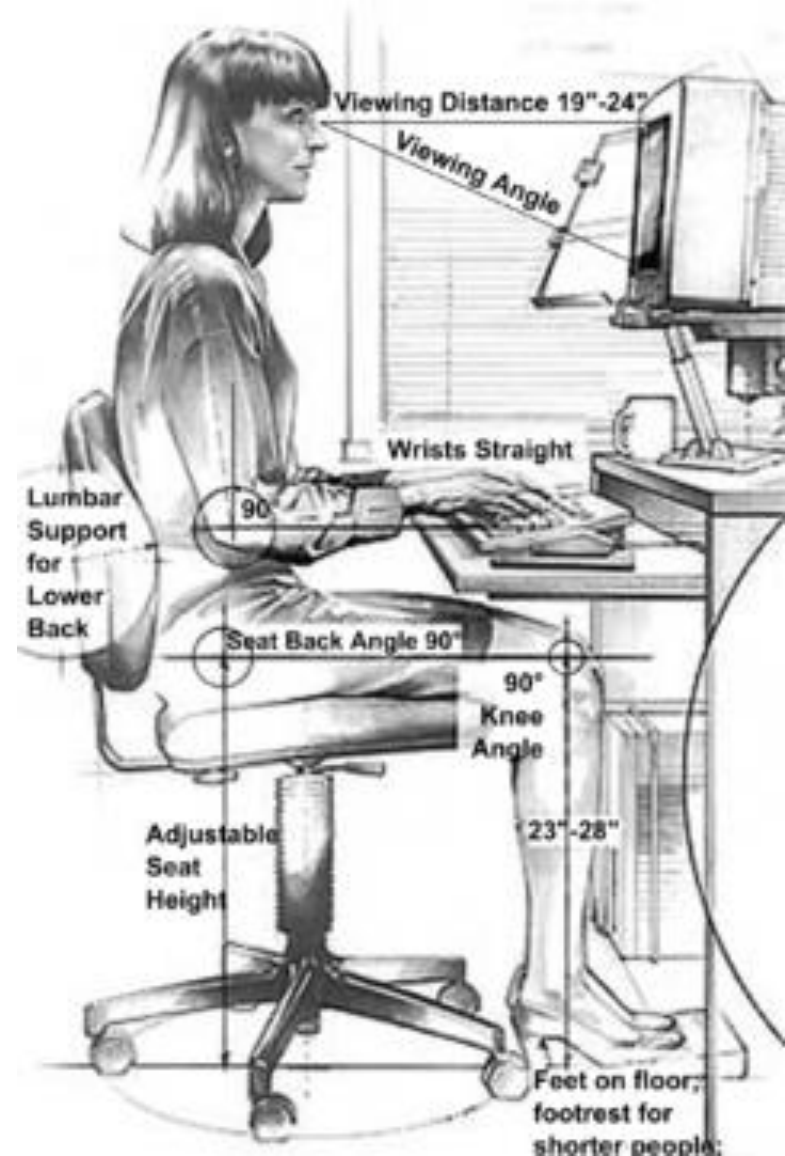
- Ergonomics comprise three main fields of research: physical, cognitive and organizational ergonomics.

Physical ergonomics

- Physical ergonomics is concerned with human anatomy, and some of the anthropometric, physiological and bio mechanical characteristics as they relate to physical activity.
- Physical ergonomic principles have been widely used in the [design](#) of both consumer and industrial products.
- Risk factors such as localized mechanical pressures, force and posture in a sedentary office environment lead to injuries attributed to an occupational environment
- Physical ergonomics is important to those diagnosed with physiological ailments or disorders such as [arthritis](#) (both chronic and temporary) or [carpal tunnel syndrome](#).
- Pressure that is insignificant or imperceptible to those unaffected by these disorders may be very painful, or render a device unusable, for those who are.
- Many ergonomically designed products are also used or recommended to treat or prevent such disorders, and to treat pressure-related [chronic pain](#).

...Physical ergonomics

- Physical ergonomics: the science of designing user interaction with equipment and workplaces to fit the user.



Cognitive ergonomics

- Cognitive ergonomics is concerned with mental processes, such as perception, memory, reasoning, and motor response, as they affect interactions among humans and other elements of a system.
- (Relevant topics include mental workload, decision-making, skilled performance, human reliability, work stress and training as these may relate to human-system and [Human-Computer Interaction](#) design.)
- Studies show a correlation between the time one spends sedentary and their cognitive function such as lowered mood and depression.

Organizational ergonomics

- Organizational ergonomics is concerned with the optimization of socio-technical systems, including their organizational structures, policies, and processes.
- Relevant topics include communication, crew resource management, work design, [work systems](#), design of working times, teamwork, [participatory design](#), community ergonomics, cooperative work, new work programs, virtual organizations, telework, and quality management.

Ergonomics and HCI

- **Ergonomics** and **HCI** are related to each other because **ergonomics** is the working environment the end user is operating within and **HCI** is the interaction the end user actually has with the computer system.